# Exit Ticket · reflect and retrieve

## Exit Ticket

An undirected edge normally changes two entries in an adjacency matrix.

## Use key vocabulary from memory

1. Explain graph in your own words.

2. Explain vertex in your own words.

3. Explain edge in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 021 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Graphs: directed and undirected

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Which structure is LIFO?

### Q2

Which structure is FIFO?

### Q3

In an array-based linked list, what does the next pointer store?

### Q4

In an undirected graph, if A is connected to B, is B also connected to A?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Create — CODE:
int[,] graph = new int[3,3]; — Three vertices (A, B and C) produce a 3 by 3 matrix: one row and one column per vertex. The four-vertex 4×4 example comes later in You Do.

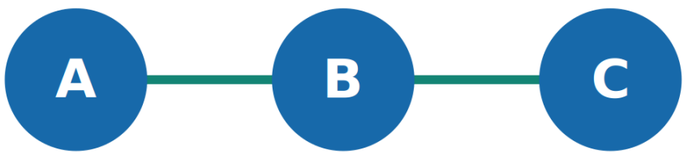
## Core idea 2

Add undirected edge — CODE:
graph[a,b] = 1;
graph[b,a] = 1; — The matrix is symmetrical for an undirected connection.

## Core idea 3

Add directed edge — CODE:
graph[a,b] = 1; — Only A→B is stored.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 2

Show direction matters
Create a 2×2 directed matrix with 0→1 only, then output both directional entries.

**Question 2 · original source diagram**




Check that you have used the technical terms and explained your decisions.

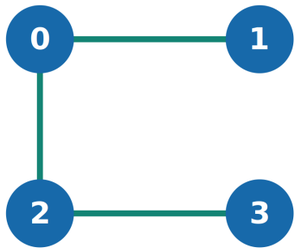
# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Create an undirected graph
Create a 4×4 matrix and add 0—1, 0—2 and 2—3.

**Question 1 · original source diagram**




Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Core idea 4

Remove and inspect — CODE:
graph[a,b] = 0;

for (int v = 0; v < graph.GetLength(1); v++)
{
if (graph[a,v] == 1)
Console.WriteLine(v);
} — Clearing removes an edge; scanning a row finds neighbours.

## Vocabulary · use the correct technical terms

graph — Vertices connected by edges.

vertex — A point/node in a graph.

edge — A connection between vertices.

directed — Edges have a direction.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Build A—B—C

## Follow the worked method

1. Make matrix — CODE:
int[,] graph = new int[3,3];

2. B—C — CODE:
graph[1,2] = 1;
graph[2,1] = 1;

3. Read neighbours of B — CODE:
row 1 = 1 0 1
B connects to A and C

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Fill the same code
CODE:
graph[0,2] = ?;
graph[2,0] = ?;

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.